8 **(a)** State Newton's law of gravitation.

……

……

……

…… [2]

(b) Earth rotates about its axis with a period of 24 hours. Assume Earth has a uniform density.
Radius of Earth = 6.37×10^6 m
Mass of Earth = 5.97×10^{24} kg

(i) Calculate the centripetal acceleration of a man standing at Earth's equator.

centripetal acceleration = …… m s^{-2} [2]

(ii) Hence, calculate the acceleration of free fall at the equator of Earth, and explain why this value may be different from that at the poles.

……

……

…… [4]

(c) The International Space Station (ISS) revolves the Earth in a circular orbit at a height of just 408 km above Earth's surface.

(i) Show that the period of the ISS is 1.5 hours.

[2]

(ii) For a circular orbit, the radius, r, and period, T, are related by the relationship

$$T^2 \propto r^3$$

This result is also known as Kepler's Third Law.

A student noted that a point on Earth's equator rotates with a period of 24 hours but the ISS in **(c)(i)** orbits with a period of just 1.5 hour.

Comment the apparent discrepancy between the student's observation and Kepler's Third Law.

……

……

……

……

……

[2]

(iii) An astronaut on the ISS deduced that he must be weightless since he was floating. Comment on his deduction.

……

……

……

……

[2]

(d) The table shows data relating the Moon orbiting the Earth and an electron orbiting the nucleus of a hydrogen atom.

	Moon orbiting Earth	electron orbiting nucleus
mass / kg	7×10^{22}	9×10^{-31}
speed / m s^{-1}	1×10^{3}	2×10^{7}
orbital radius / m	4×10^{8}	5×10^{-11}

Fig. 8.1

(i) Using the data from Fig 8.1, determine

1. the de Broglie wavelength of the Moon orbiting the Earth,

wavelength = …….. m

2. the de Broglie wavelength of the electron orbiting the nucleus.

wavelength = …….. m [3]

(ii) Hence, explain why it is reasonable to treat the Moon in orbit around the Earth as a "particle" but it is not reasonable to treat the electron in orbit around the nucleus as a "particle".

……

……

……

……

…… [3]